Earth-Observation Data Processing: Second report - ISM Test

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The Instrument Module simulates the payload acquisition process by transforming Top-of-Atmosphere (toa) radiances through three main stages to generate raw digitized imagery. In the Optical Stage, incoming radiances are spectrally filtered using the Instrument Spectral Response Function, converted to irradiances based on telescope geometry, and spatially blurred via a system Modulation Transfer Function that accounts for diffraction, optical aberrations, defocusing, detector footprint, and motion smearing. The Detection Stage models the physical conversion of this irradiance into photons and subsequently into electrons based on the sensor's quantum efficiency, while simultaneously introducing realistic sensor non-idealities such as Photo Response Non-Uniformity (PRNU), thermal Dark Signal, and signal attenuation from dead or bad pixels. Finally, the Video Chain Stage simulates the electronic read-out by converting the accumulated electron charges into voltage through specific amplification gains, which are then digitized into 12-bit Digital Numbers to produce the final output.

The results obtained were validated by comparing them with the outputs provided by the teacher.

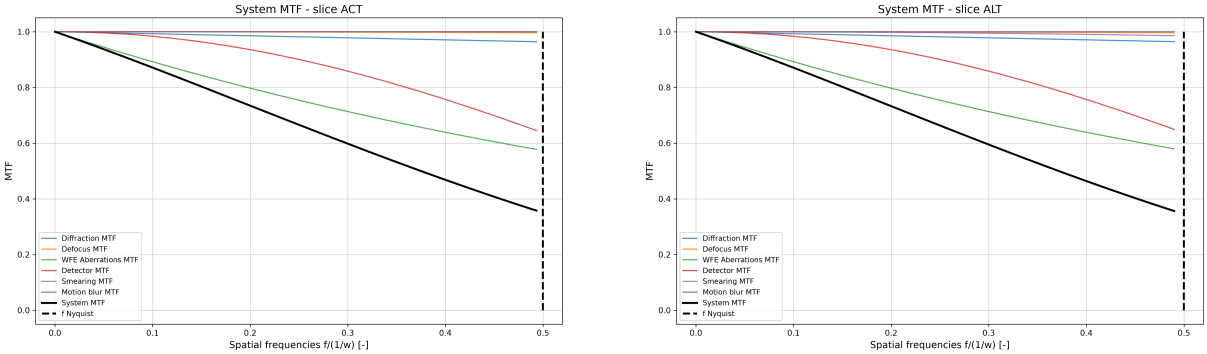


Figure 1: System MTF and MTF components in ACT and ALT directions.

The system's spatial resolution in both the ACT and ALT directions is primarily limited by the Wavefront Error (WFE) Aberrations MTF and the Detector MTF. In the provided plots, these two components represent the lowest curves, driving the overall System MTF down. The remaining simulated effects (diffraction, defocus, smearing, and motion blur) stay very close to 1.0 at the Nyquist limit, meaning they introduce negligible attenuation to the image detail.

- WFE Aberrations MTF (Green Curve): This represents the optical imperfections within the telescope. It models the degradation that occurs because the instrument's physical optics cannot perfectly focus all incoming parallel wavefronts into a single ideal point, causing the image to blur spatially across the focal plane.
- Detector MTF (Red Curve): This accounts for the finite physical area of the individual detector elements. Because a pixel integrates incoming irradiance over its entire rectangular footprint, it fundamentally acts as a spatial box filter. In the frequency domain, this geometric integration translates to a sinc function that naturally attenuates higher spatial frequencies.

Table 1 summarizes the sequential stages followed in the conversion of the TOA radiances, outlining the physical units at each step and the mathematical conversion factors applied to transition from the optical input to the final digitized output.

TOA Stage	**Unit**	**Conversion Factor**
isrf	$\mathrm{mW}/(\mathrm{m}^2 \cdot \mathrm{sr})$	$T_r \cdot \frac{\pi}{4} \cdot \left(\frac{D}{f}\right)^2$ [sr]
toa_irr	mW/m^2	$\frac{\mathrm{Area} \cdot t_{\mathrm{int}}/1000}{hc/\lambda}$ $\left[\frac{\mathrm{ph}}{\mathrm{mW}/\mathrm{m}^2}\right]$ (for VNIR-0)
toa_ph	ph	QE $[\mathrm{e}^-/\mathrm{ph}]$
toa_e	e^-	$\mathrm{e2V} = \mathrm{OCF} \cdot \mathrm{Gain}$ $[\mathrm{V}/\mathrm{e}^-]$
toa_v	V	$\frac{2^{\mathrm{bit_depth}}-1}{V_{\max}-V_{\min}}$ [DN/V]
toa_dn	DN	—

Table 1: Unit conversions across the instrument processing stages.